

The Editor-in-Chief
Information Sciences
Elsevier

Dear Editor-in-Chief,

We respectfully submit our manuscript, “**Limits of Fibonacci Index-Gap Coding: Complete Stream Costs, Compression Bounds, and Structural Controls,**” for consideration as a Research Article in *Information Sciences*.

Scope fit. The journal explicitly includes information theory, algorithm design, computer-system evaluation, and data compression. The manuscript studies a lossless positional representation under exact stream accounting: every model, delimiter, selector, length field, and padding bit required for reconstruction is measured. It combines analytical bounds, algorithm specification, compiled evaluation, and structural controls.

Principal results.

- **Exact accounting:** A serialized format and a necessary-and-sufficient gain inequality determine whether a complete positional stream is shorter than its input, without treating metadata as free. Here, complete cost is defined within the serialized stream boundary.
- **Formal and empirical validation:** The transform is proved bijective; its sufficient term bound is calibrated on Canterbury and Silesia; generic 2048-bit integers are shown analytically and empirically to lie far above that bound; and a compiled implementation verifies every decoding round trip on complete corpora and the 100 MB enwik8 benchmark.
- **Structural controls:** Lucas indices, ordinary binary set-bit positions, zero-block bitmaps, and boundary-zero trimming separate number-system effects from positional sparsity and simpler zero structure.
- **Reproducibility:** The public package provides the format implementation, tests, configurations, result tables, figure-generation scripts, and exact seeds.

Evidence and interpretation. A full-corpus sweep over five block sizes places the best FISA ratios at 92.15% for Canterbury and 98.78% for Silesia, but only one and two files compress, respectively. Matched zero-bitmap and zero-trim controls are smaller, and established codecs win every paired file comparison. The manuscript treats this outcome as a limit of the Fibonacci construction, not as evidence for a competitive general-purpose codec.

Relevance to readers. The framework is applicable to information-processing methods that introduce shared dictionaries, codebooks, model parameters, block selectors, or other amortized side information. It offers researchers and practitioners a reproducible procedure for separating payload reduction from complete, decodable system cost.

Author declarations.

- **Originality and exclusive submission:** The manuscript is original, has not been published previously, and is not under consideration elsewhere.
- **Author approval:** All named authors have approved the manuscript, the authorship order, and the stated contributions.
- **Competing interests and funding:** The authors declare no competing interests. This research received no external funding.
- **Ethical compliance:** The study involves no human participants, animals, personal data, or clinical interventions; it uses public benchmark corpora, synthetic data, and offline computational experiments.

We believe that the combination of information-theoretic analysis, exact format accounting, compression experiments, and reproducible negative evidence fits the theory-and-practice scope of *Information Sciences*. Thank you for considering this submission.

Yours sincerely,

Dr. Stéphane Gaël R. Ekodeck (Corresponding Author)

ORCID: 0000-0002-8094-8832

LIRIMA, TEAM GRIMCAPE, University of Yaounde I, Yaounde, Cameroon

stephane-gael.ekodeck@facsciences-uy1.cm

Co-authors: Hervé Talé Kalachi, Serge Alain Ebélé, Norbert Djong Wang, and Chantal Marguerite Mveh-Abia.